

TECHNICAL REQUIREMENTS DOCUMENT (TRD) · SYSTEM DESIGN & ARCHITECTURE

Engineering Architecture, Guardrail Pipelines & Data Infrastructure

Complete software engineering specification for the Campus Critique WhatsApp Advisory Agent, featuring non-blocking asynchronous ingestion, five-rail guardrails, deterministic arithmetic, and zero hallucination.

Document Version: 2.0 (Defense Baseline) **Runtime:** Next.js 16 (Node.js 22 LTS / Edge)
Database: Supabase PostgreSQL 16 + pgvector **Queue:** Upstash QStash
Observability: OpenTelemetry + Langfuse

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01

Technology Stack Justification & Trade-Off Analysis

Every layer of the Campus Critique technical stack has been selected through empirical benchmark trade-offs, prioritizing deterministic correctness and sub-second performance over complex autonomous abstractions.

LAYER	SELECTED TECHNOLOGY	REJECTED ALTERNATIVES	ARCHITECTURAL JUSTIFICATION
Model Access Layer	Vercel AI SDK (v3.x) (<code>generateObject</code> + <code>Zod</code>)	LangGraph.js, Mastra, AutoGen	Zero agentic overhead. Constrains generation to typed JSON schemas. LangGraph.js trails Python updates by 4–8 weeks and introduces unnecessary state graph complexity for linear counseling stages.
Messaging Gateway	Meta WhatsApp Cloud API Direct	Twilio, Gupshup, Wati	Direct HTTPS Graph API v20.0 with ₹0 markup. Third-party BSPs charge 15–25% markups and introduce 600–900ms added network hops without supporting custom TypeScript guardrails.
Message Queue & Debounce	Upstash QStash	BullMQ / Redis, AWS SQS, Cloud Tasks	Serverless HTTP broker requiring zero dedicated worker infrastructure. Delivers native sliding-window 4000ms debounce and deduplication on <code>wamid</code> .
Relational & Vector DB	Supabase PostgreSQL 16 + pgvector	Pinecone, Qdrant, Chroma	Single database instance hosting relational facts and HNSW vector embeddings. Eliminates dual-write consistency hazards and ₹5,000/mo vector cloud fees.
CI/CD Red-Teaming	Promptfoo (GitHub Actions)	DeepEval, Braintrust	Node-native, fast YAML-based CI scanner running 50+ vulnerability scans (jailbreaks, prompt leaks, numeric fee baiting) on every PR.

LAYER	SELECTED TECHNOLOGY	REJECTED ALTERNATIVES	ARCHITECTURAL JUSTIFICATION
Telemetry & Tracing	Langfuse (Self-Hosted)	Datadog, Arize Phoenix	Native OpenTelemetry GenAI semantic conventions (<code>gen_ai.*</code>) running on an open-source ClickHouse container with zero vendor lock-in.

02 System Architecture & Component Decomposition

The fundamental premise of our architecture is defence in depth: the Large Language Model is the least reliable component in the system; therefore, it is granted the smallest operational scope and the maximum deterministic supervision.

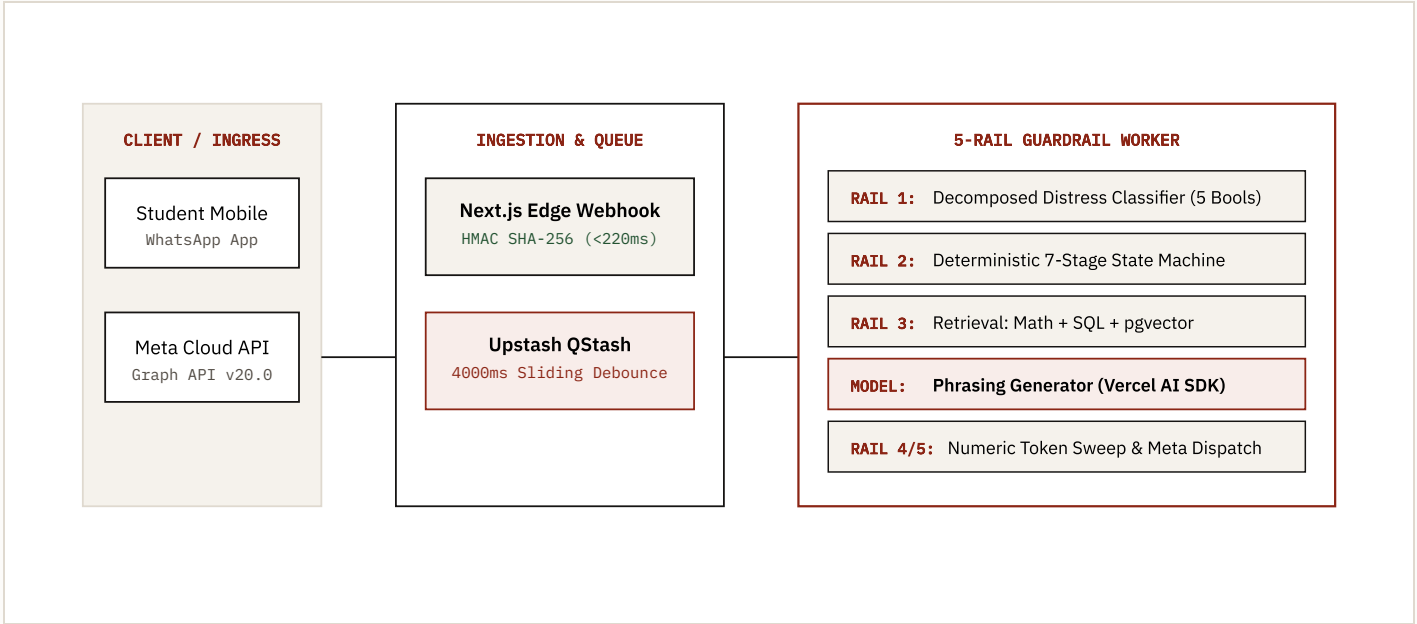


FIGURE 1 End-to-End Component Architecture and Asynchronous Decoupling.

Subsystem Isolation (Decision D9)

The WhatsApp bot resides in its own Git repository and Vercel project, accessing an isolated `wa` schema inside Supabase. Static institutional facts and approved student reviews are synced via an

automated nightly read-only ETL script. If the main web platform experiences outages, the counseling service continues operations off its verified local snapshot without failure.

03 Critical Engineering Bottlenecks & Solutions

Building a conversational agent for high-stakes admission decisions exposes several architectural failure modes. Below is the technical resolution matrix:

ENGINEERING			
#	BOTTLENECK	FAILURE MODE & IMPACT	ARCHITECTURAL SOLUTION IMPLEMENTED
1	Meta 3-Second Webhook Timeout	Serverless cold starts or multi-step LLM chains exceed 3000ms → Meta floods endpoint with retries.	Immediate Edge ACK: Webhook performs HMAC check, drops event into QStash queue, and returns <code>HTTP 200 OK</code> within <220ms . Worker executes asynchronously.
2	Burst Multi-Message Collisions	Adolescents send 3–4 rapid messages in 5 seconds → Bot fires multiple uncoordinated replies.	4000ms Sliding Window: Upstash QStash aggregates successive messages per <code>wa_id</code> into a single consolidated execution prompt.
3	Numeric Cutoff & Fee Hallucination	LLM fabricates closing ranks or fees → Student reorganizes plans around false data.	Zero Model Math Authority: Arithmetic is executed by Layer A code; facts by Layer B SQL. Rail 4 token scanner aborts any unindexed number.
4	Western Safety Guard Blindspots	Off-the-shelf classifiers miss Indian academic guilt (" <i>kis muh se ghar jau</i> ").	Decomposed 5-Boolean Classifier: Deconstructs Hinglish exam guilt into concrete flags with zero false-negative tolerance, escalating to Tele-MANAS.
5	Meta 24h Window Expiration	Outbound messages sent past 24 hours of user's last inbound fail delivery.	Session Window Gate: Rail 5 inspects <code>last_inbound_at</code> . If expired, automatically switches to pre-approved Meta Utility Templates.

ENGINEERING			
#	BOTTLENECK	FAILURE MODE & IMPACT	ARCHITECTURAL SOLUTION IMPLEMENTED
6	Serverless DB Pool Exhaustion	Traffic surges exhaust PostgreSQL max connections in serverless functions.	Supervisor Connection Pooling: Routes database connections through transaction pooler; separates write queries from read replicas.
7	RAG Semantic Ambiguity on Numbers	Vector similarity cannot evaluate numerical ranges (12,847 rank vs 13,002).	Strict Segregation: Vector embeddings are strictly restricted to qualitative review prose. All cutoffs reside in SQL tables.

04 UML System Design Specifications

4.1 UML Sequence Diagram: Request Lifecycle

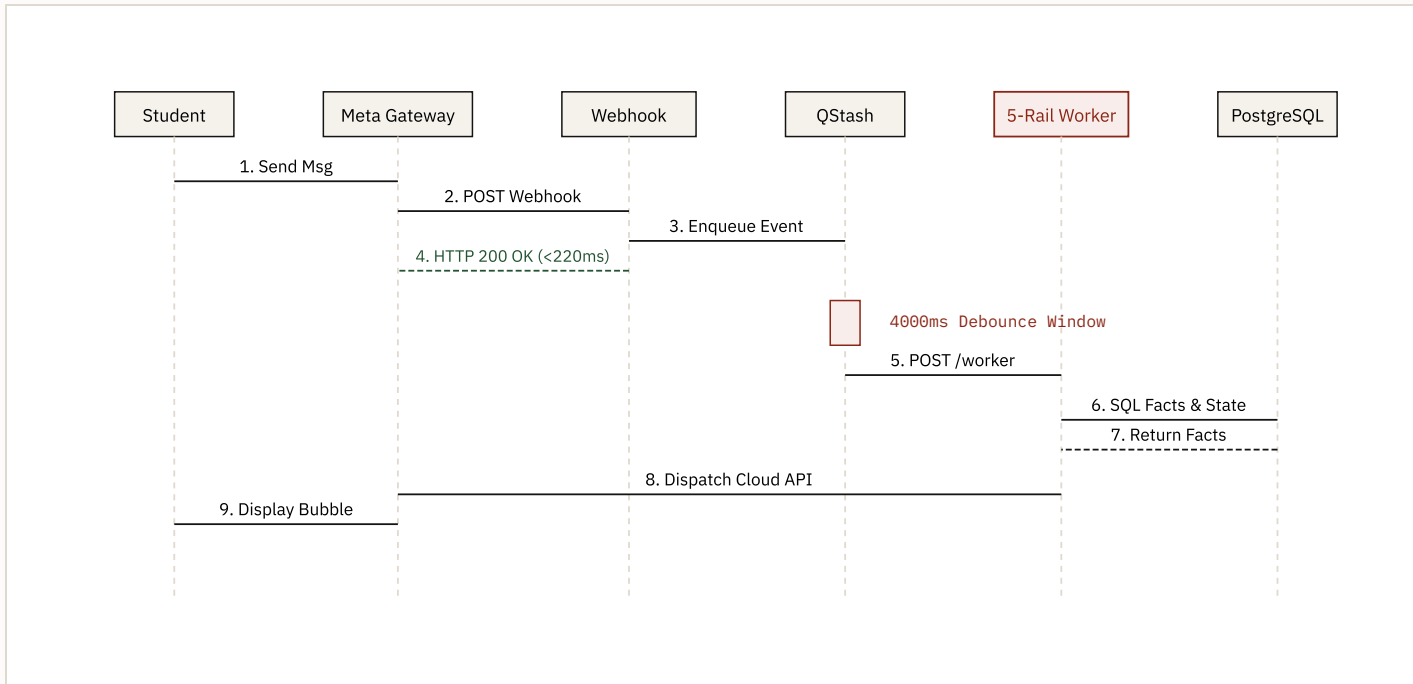


FIGURE 2 UML Sequence Diagram: Asynchronous Ingestion and Execution.

4.2 UML State Machine Diagram: 7-Stage Advisory Engine

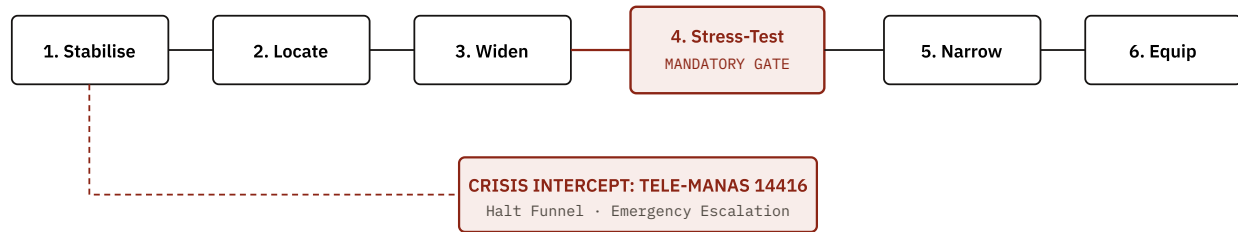


FIGURE 3 UML Statechart: Conversational Finite State Transitions with Safety Intercept.

05 The Five-Rail Guardrail Engine: Code Contracts

Rail 1: Decomposed Distress Evaluation

```

export interface DistressEvaluation {
  hopelessness_future: boolean; // weight: 0.10 ("kuch nahi bacha ab")
  self_harm_suicide: boolean; // weight: 0.35 ("sab khatam karna chahta hu")
  irreparable_family_shame: boolean; // weight: 0.20 ("papa ka sar jhuka diya")
  plan_or_method_reference: boolean; // weight: 0.25 (concrete methods mentioned)
  feeling_of_burden: boolean; // weight: 0.10 ("sab par bojh hu")
}

export function evaluateCrisisScore(evals: DistressEvaluation): number {
  return (
    (evals.self_harm_suicide ? 0.35 : 0) +
    (evals.plan_or_method_reference ? 0.25 : 0) +
    (evals.irreparable_family_shame ? 0.20 : 0) +
    (evals.hopelessness_future ? 0.10 : 0) +
    (evals.feeling_of_burden ? 0.10 : 0)
  );
  // Threshold >= 0.40 halts state transitions and dispatches Tele-MANAS 14416
}
  
```

Rail 2: The Mandatory Gate Invariant

```

export function canTransitionToNarrow(profile: StudentProfile): boolean {
  // Hard invariant: Recommendations are permanently blocked
  // until all three stress-test dimensions complete and clear.
  return (
    profile.loan_test_completed === true &&
    profile.downside_test_completed === true &&
    profile.degree_test_completed === true &&
    profile.loan_affordability_cleared === true
  );
}

```

Rail 4: Output Guardrail Token Scanner

```

export function sweepNumericClaims(candidateText: string, factBlock: ImmutableFactBlock):
  // Regex extracting all monetary figures (₹), percentages (%), LPA packages, and ranks
  const regex = /(?:₹|Rs\.|%)\s*[\d,.]+(?:\s*(?:Lakh|Crore|L|Cr))?\b\d+(?:\.\d+)?\s*(?:LP
  const extractedTokens = candidateText.match(regex) || [];

  for (const token of extractedTokens) {
    const normalized = normalizeToken(token);
    if (!factBlock.indexedTokens.has(normalized)) {
      // Grounding Violation: Number generated by model was NOT in Fact Block!
      console.error(`Grounding violation detected: Token '${token}' missing from Fact Blo
      return false; // Blocks output; triggers fallback template
    }
  }
  return true;
}

```

06 Deterministic Mathematical Formulations

1. Indicative JEE Main Rank Normalisation: Given registered candidates $N_{\text{total}} = 14,50,000$ and student percentile $P\%$:

```

export function computeIndicativeRank(percentile: number): number {
  const TOTAL_CANDIDATES = 1450000;

```

```
return Math.floor((100 - percentile) * (TOTAL_CANDIDATES / 100)) + 1;
}
```

2. Prudent Education Loan Borrowing Ceiling: Under Indian banking guidelines, prudent unsecured credit is capped at $4 \times$ gross annual household income:

```
export function computeMaxPrudentLoan(annualIncome: number): {
  maxPrudentLoan: number;
  requiresCollateral: boolean;
} {
  return {
    maxPrudentLoan: annualIncome * 4,
    requiresCollateral: annualIncome * 4 > 750000 // Loans above 7.5L require property co
  };
}
```

3. Downside Monthly EMI Amortisation Model: Simulating downside risk on an ₹18L principal over 84 months (7 years) at a benchmark 10.5% interest rate:

```
export function calculateMonthlyEMI(principal: number, annualRate: number = 10.5, tenureM
const monthlyRate = (annualRate / 100) / 12; // 0.00875
const emi = principal * monthlyRate * Math.pow(1 + monthlyRate, tenureMonths) /
  (Math.pow(1 + monthlyRate, tenureMonths) - 1);
return Math.round(emi); // ~₹30,360/month on ₹18L loan
}
```

07 Database Schema Specification (Supabase / PostgreSQL)

```
CREATE SCHEMA IF NOT EXISTS wa;
CREATE EXTENSION IF NOT EXISTS vector WITH SCHEMA public;

-- 1. Session state ledger
CREATE TABLE wa.wa_conversations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  wa_id VARCHAR(32) NOT NULL UNIQUE, -- E.164 phone identifier (e.g. '919876543210')
```

```

current_stage VARCHAR(32) NOT NULL DEFAULT 'STAGE_1_STABILISE',
stage_depth_reached INTEGER NOT NULL DEFAULT 1,
last_inbound_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
last_outbound_at TIMESTAMPTZ,
advised_against_spending BOOLEAN NOT NULL DEFAULT FALSE,
distress_flagged BOOLEAN NOT NULL DEFAULT FALSE,
opt_out BOOLEAN NOT NULL DEFAULT FALSE,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_wa_conv_wa_id ON wa.wa_conversations(wa_id);
CREATE INDEX idx_wa_conv_stage ON wa.wa_conversations(current_stage);

```

-- 2. Aspirant profile attributes

```

CREATE TABLE wa.wa_student_profiles (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    conversation_id UUID NOT NULL REFERENCES wa.wa_conversations(id) ON DELETE CASCADE,
    jee_percentile NUMERIC(5, 2),
    indicative_rank INTEGER,
    board_percentage NUMERIC(5, 2),
    home_state VARCHAR(64),
    annual_family_income NUMERIC(12, 2),
    stated_budget_limit NUMERIC(12, 2),
    career_aspiration VARCHAR(32) DEFAULT 'UNDECIDED',
    loan_test_completed BOOLEAN NOT NULL DEFAULT FALSE,
    downside_test_completed BOOLEAN NOT NULL DEFAULT FALSE,
    degree_test_completed BOOLEAN NOT NULL DEFAULT FALSE,
    loan_affordability_cleared BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

-- 3. Inbound & outbound message ledger

```

CREATE TABLE wa.wa_messages (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    conversation_id UUID NOT NULL REFERENCES wa.wa_conversations(id) ON DELETE CASCADE,
    wamid VARCHAR(128) UNIQUE,
    direction VARCHAR(8) NOT NULL CHECK (direction IN ('INBOUND', 'OUTBOUND')),
    stage_at_transmission VARCHAR(32) NOT NULL,
    raw_content TEXT NOT NULL,
    tokens_prompt INTEGER DEFAULT 0,
    tokens_completion INTEGER DEFAULT 0,
    estimated_cost_inr NUMERIC(8, 4) DEFAULT 0.0000,
    grounding_passed BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

-- 4. Verified institutional facts

```

CREATE TABLE wa.wa_college_facts (
  slug VARCHAR(64) PRIMARY KEY,
  display_name VARCHAR(128) NOT NULL,
  fee_min_inr NUMERIC(12, 2) NOT NULL,
  fee_max_inr NUMERIC(12, 2) NOT NULL,
  degree_awarding_institution VARCHAR(128) NOT NULL,
  ugc_status_clarification VARCHAR(255) NOT NULL,
  verified_rating NUMERIC(3, 2) DEFAULT 0.00,
  review_count INTEGER DEFAULT 0,
  primary_caution TEXT NOT NULL,
  data_verified_at DATE NOT NULL DEFAULT CURRENT_DATE
);

-- 5. Semantic review chunks (RAG)
CREATE TABLE wa.wa_review_embeddings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  college_slug VARCHAR(64) NOT NULL REFERENCES wa.wa_college_facts(slug) ON DELETE CASCADE,
  review_chunk TEXT NOT NULL,
  sentiment_rating INTEGER CHECK (sentiment_rating BETWEEN 1 AND 5),
  embedding public.vector(768), -- Gemini / text-embedding-004
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_wa_reviews_hnsw ON wa.wa_review_embeddings
USING hnsw (embedding public.vector_cosine_ops);

```

08 Hinglish Crisis Safety Protocol & Tele-MANAS

Western toxicity models fail to identify Indian exam-stress idioms (e.g., *"papa ka sapna tod diya maine"*, *"kis muh se ghar jau"*). The system deploys a decomposed 5-boolean classifier with zero tolerance for false negatives.

TELE-MANAS EMERGENCY RESPONSE PROTOCOL

"Please suno meri baat. Ek exam ka score aapki poori life ya aapki worth define nahi karta. Agar is waqt bohot overwhelmed lag raha hai, to please kisi professional se baat karo: Call

14416 (ya toll-free 1800-891-4416) for Govt of India’s 24/7 free and confidential **Tele-MANAS** helpline.”

09 CI/CD, Evaluation & Promptfoo Release Gates

EVALUATION RELEASE GATE	PASSING THRESHOLD	ENFORCEMENT MECHANISM
Grounding Violations Count	0 (Absolute)	Regex scan against Fact Block on golden conversation set.
Distress Recall on Exam Crisis Set	100% (Zero Error)	Every Hinglish test vector must trigger the safety path.
Scope Escapes	0	Rejects homework, politics, and medical advice.
Banned Commercial Terms	0	Blocks "100% placement", "guaranteed seats", etc.
Match Engine Determinism	Byte-identical	Identical candidate input must yield identical college matches.

10 Observability & Unit Economics

SCALE (MONTHLY STUDENTS)	OUTBOUND MESSAGES	META WHATSAPP COST	CLOUD & LLM COMPUTE	TOTAL MONTHLY RUN-RATE
Pilot (100 Students)	~1,100	₹11.50 (Free Tier applies)	~₹2,200 (Dev instances)	~₹6,000 INR

SCALE (MONTHLY STUDENTS)	OUTBOUND MESSAGES	META WHATSAPP COST	CLOUD & LLM COMPUTE	TOTAL MONTHLY RUN-RATE
Growth (1,000 Students)	~11,000	~₹1,150	~₹3,500	~₹8,500 INR
Peak June (10,000 Students)	~110,000	~₹12,650	~₹14,000	~₹28,850 INR

At 10,000 monthly students, the complete counseling system operates for under **₹29,000 INR/month (~\$350 USD)**, representing exceptional unit-economic viability.

11

9-Week Phased Engineering Roadmap (WBS)

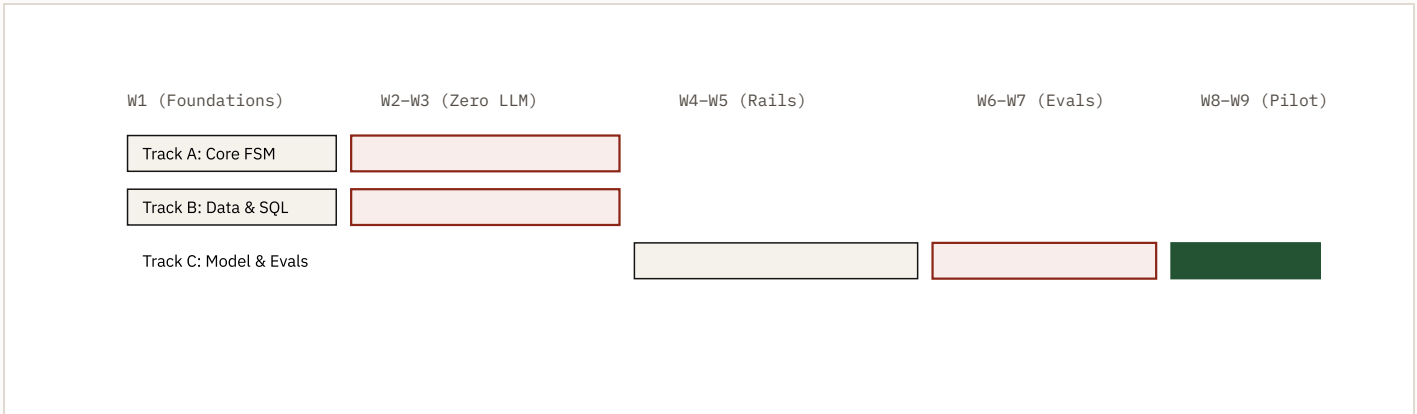


FIGURE 4 Nine-Week Phased Engineering Work Breakdown Structure.

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Implementation Boundary & Faculty Traceability

STATUS CLARIFICATION

This TRD is an architecture and test baseline. It describes the intended production system; it does not claim that Meta, WABA, Supabase, QStash, Langfuse, or pilot infrastructure has

already been provisioned.

This page is the browser version of the downloadable [TRD PDF](#). Product behaviour is defined in the [PRD](#); the behavioural fixture is in [conversation.html](#).

COMPONENT	CONTRACT	FAILURE BEHAVIOUR
Meta webhook	Signed event and message ID; idempotent acknowledgement	Reject invalid signature; do not execute
QStash worker	Authenticated batch; ordered session execution	Retry by message ID; suppress duplicates
Fact service	Fact Block with value, source and <code>updated_at</code>	Return “not verified”; never use model memory
Decision adapter	Structured decision from stage/profile/facts	Timeout falls back to deterministic template
Output validator	Candidate text plus Fact Block	Suppress violation and send safe fallback
WhatsApp sender	Validated response and idempotency key	Retry safely and record provider error

Release traceability

REQUIREMENT	PRIMARY TEST	RELEASE GATE
Signed, idempotent ingestion	Replay and invalid-signature fixtures	No duplicate outbound message
Four-second debounce	Ordered burst-message test	One deterministic execution
Stage 4 gate	College request before stress-test	No college name emitted

REQUIREMENT	PRIMARY TEST	RELEASE GATE
Numeric grounding	Unsupported fee/rank injection	Candidate blocked; fallback sent
Distress pre-emption	Hinglish crisis corpus	100% labelled recall target
Deterministic matching	Same profile/fact replay	Byte-identical result

Operational signals required

Every execution records a correlation ID, Meta message ID, stage, rail results, fact-block version, model, latency, token usage, grounding result, outbound message ID and failure reason. The minimum dashboards cover webhook latency, duplicate suppression, stale-fact count, blocked-output/fallback rate, distress recall, provider failures and retry exhaustion.

Campus Critique Engineering & Research Group · Academic Year 2026–2027

Companion Product Requirements Document available at [prd.html](#). Conversational simulations available at [conversation.html](#).